

A thousand-state optoelectronic memory for high-precision spatiotemporal encoding

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Conventional photodiodes can distinguish subtle illumination variations, but fail to retain information. Conversely, optoelectronic memories can store light information but encounter difficulties in multiple-state storage due to noise from the deexcitation recombination of the photogenerated carriers. Here we develop an optoelectronic memory that can generate a multilevel response to light stimuli and retain the multi-states in a nonvolatile manner. In the designed type-III heterojunction device, electrons flow from high-Fermi-level (E_F) p-type amorphous carbon to low- E_F n-type TiO_x , generating an unusual built-in electric field that enhances majority carrier transport. Photogenerated electrons from the oxygen vacancies in TiO_x quickly combine with holes from amorphous carbon, effectively reducing deexcitation recombination with charged vacancies and associated noise. The resulting device achieves 1,024 distinguishable optoelectronic memory states without any denoising process, enabling high-precision spatiotemporal information encoding. The thousands of states in optoelectronic memory allows for the emulation raptor vision to perception fast-moving objects.

Bioinspired in-sensor vision processing highly demands devices that can store multiple distinct optoelectronic memory states for accurate perception of complex and delicate motions^{1–5}. Conventional photodiodes, however, are unable to retain light stimuli owing to their transient response. While optoelectronic memories can preserve the information after the removal of light stimuli, they face challenges such as noise from deexcitation recombination and the relaxation of photogenerated electrons and holes^{6–8}. These factors lead to conductance fluctuations, making optoelectronic memory states indistinguishable and limiting the storage of multiple states.

For an analog resistive switching memory, the intrinsic tunable conductance range allows for an unlimited number of states, because the conductance gap (ΔG) between adjacent states can be infinitely divided⁹. However, the adjacent states of optoelectronic memory can overlap due to spontaneous electron/hole recombination and relaxation^{10–12}. When ΔG is small, the neighboring conductance states

merge into one state because of noise interference, compromising the storage capacity of the device. In most optoelectronic memory devices, the deexcitation recombination of photogenerated carriers as major noise causes photoconductance relaxation and overlap^{13–15}, thus limiting the encoding precision. For electrically driven resistive switching memories, random telegraph noise (RTN) associated with defect dynamics causes fluctuations at each conductance state¹⁶. After suppressing RTN from hundreds of nanoamperes to tens of nanoamperes by applying small voltages, Yang et al. demonstrated thousands of conductance states with small ΔG of $2.0 \mu\text{S}$ ¹⁷. However, the optoelectronic response involves multiple physical processes, including photocarrier generation, transport, and recombination. Defects can be used to alter the decay kinetics of photogenerated carriers^{18–20}, leading to persistent photoconductivity for information storage. The noise from carrier dynamics of defects can undoubtedly impair precise conductance control in optoelectronic memory devices²¹. Meanwhile,

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the requirements for real-time response in optoelectronic perception also preclude the possibility of employing a post-programming denoising scheme. Consequently, it is essential to explore alternative strategies for multilevel optoelectronic memories that can effectively mitigate noise interference and achieve precise fine-grained conductance control.

In this work, we design an optoelectronic memory device with a vertical structure of fluorine-doped tin oxide (F-doped SnO_2 , FTO)/amorphous carbon (a-Carbon)/ TiO_x /Au. This type-III heterojunction generates an unusual built-in electric field that promotes majority carrier transport. Consequently, the photogenerated electrons from neutral oxygen vacancies (V_o) in n-type TiO_x can rapidly combine with the holes from p-type a-Carbon, largely suppressing the recombination of electrons and conductive oxygen vacancies (V_o^{1+} or V_o^{2+}) and resulting in conductance change of TiO_x . The designed device further enables precise tuning of light-induced conductive vacancies in TiO_x , achieving thousands of optoelectronic memory states without any denoising technique. By configuring thousands of memory states under high-frequency light stimuli, the type-III heterojunction optoelectronic memory can emulate raptor to precisely fuse spatiotemporal vision information for perception complex fast-moving objects.

Results and discussion

Mechanisms of type-III heterojunction optoelectronic memory

In a conventional photodiode, electrons and holes quickly recombine after removing light stimuli, thus only supplying a transient photoresponse and failing to store information after light stimuli (left of Fig. 1a and Supplementary Fig. 1). An optoelectronic memory device^{22–24} can store information after light stimuli, such as photogenerated electrons stored in oxygen vacancies. For the designed optoelectronic memory based on the a-Carbon/ TiO_x heterojunction, it relies on the carrier recombination at broadened space charge region (positive bias condition) to finely modulate the concentration of the charged oxygen vacancies, thus building thousands of optoelectronic memory states (right of Fig. 1a). Figure 1b schematically compares the photoresponse of the optoelectronic memory with a conventional photodiode. The photodiode can respond fast to light stimuli, but fails to build optoelectronic memory states. In contrast, optoelectronic memory devices show multiple memory states through subtly modulating the ΔG . Within intrinsic conductance range, distinguishable state number n is governed by the ΔG , which is primarily dominated by the change in the concentration of the storage electrons.

We fabricated an optoelectronic memory features F-doped SnO_2 /a-Carbon/ TiO_x /Au vertical structure (Fig. 1c). The valence-band maximum of p-type a-Carbon exceeds the conduction-band minimum of n-type TiO_x , creating a type-III heterojunction. For the p-type a-Carbon, the ultraviolet photoelectron spectroscopy measurement demonstrated that its Fermi level and valence-band maximum located at -3.55 eV and -4.29 eV, respectively (Supplementary Fig. 2a, b). The fitting results of the ultraviolet–visible (UV–vis) illustrated that optical band gap of the a-Carbon is 2.23 eV (Supplementary Fig. 2c). According to the valence-band maximum and the optical band gap, the conduction-band minimum is -2.06 eV. The optical band gap of 2.23 eV enables electron-hole pair generation under 405-nm ultraviolet light illumination (3.06 eV photon energy). However, the Raman spectrum illustrated that the p-type a-Carbon involves a large number of π electrons in sp^2 orbits, which are highly sensitive to 405-nm light stimuli as well as natural light (Supplementary Fig. 3).

For the n-type TiO_x , the Fermi level, valence-band maximum, conduction-band minimum, and optical band gap are -5.17 , -8.07 , -4.91 , and 3.16 eV, respectively (Supplementary Fig. 4a–c). This characteristic suggests that the electron in the valence-band of the TiO_x cannot be directly excited by the 405-nm ultraviolet light. Oxygen vacancies are introduced to form stoichiometric mismatch titanium oxide (TiO_{x-2}) during the fabrication process^{24–26}, which is verified by

the Ti_2O_3 component in the titanium oxide through X-ray photoelectron spectroscopy analysis (Supplementary Fig. 5). These vacancies establish defect levels between the valence-band maximum and conduction-band minimum, allowing electrons in these vacancy-induced levels to be excited by 405 nm ultraviolet light.

The thin films were deposited under optimized sputtering parameter, showing a low roughness ($R_a \sim 0.9$ nm) and providing a high-quality contact interface between TiO_x film and a-Carbon film (Supplementary Fig. 6). The FTO/a-Carbon forms Schottky contact ($\Phi_m > \Phi_s$) that can facilitate the photogenerated electrons to quickly transport out from the a-Carbon. The TiO_x /Au builds Ohmic contact ($\Phi_m > \Phi_s$, $\Delta\Phi \sim 0.07$ eV) that can facilitate the possible residual photogenerated electrons to transport from the TiO_x after light stimuli, increasing the stability of the optoelectronic memory states (Supplementary Fig. 7). The high-resolution transmission electron microscopy characterization shows dense contact interfaces, ensuring the photogenerated carriers with highly efficient transport (Supplementary Fig. 8). The two-terminal device structure provides downward scaling potential and integrate with waveguides that can transmit light beam with smaller spot for advanced in-sensor computing system (Supplementary Figs. 9–11). Notably, small size of the device produces high spatial resolution and the matched nanoscale light source is desirable for optical addressing. For conventional optical system within visible light range, the minimum size of light spot is in micrometer level ($1.0 - 5.0 \mu\text{m}$) due to light diffraction limit, causing the optical system, such as the spatial light modulator with $\sim 12.5 \mu\text{m}$ pixel pitch. To break through the limitation, extreme manufacturing technologies, including superstructure surface based on subwavelength nanocolumn arrays^{27,28} and nanoscale organic light-emitting diodes (OLEDs)²⁹ are developed for high-precision optical addressing. The size of a single light spot for superstructure surface nanocolumn and OLED is as low as several hundred nanometers. These technologies provide a promising chance of optical addressing for the type-III heterojunction optoelectronic memory.

Figure 1d illustrates the working principle of the designed type-III heterojunction. The Fermi level of p-type a-Carbon is higher than both the Fermi level and the conduction-band minimum of n-type TiO_x (Supplementary Fig. 12), allowing electrons from a-Carbon to fill the oxygen vacancies in TiO_x film. It produces neutral oxygen vacancies in TiO_x . As a result, the built-in electric field (E_{in}) in the type-III heterojunction is directed from p-region (a-Carbon) to n-region (TiO_x), fundamentally differing from conventional pn junction. When a positive voltage is applied on the side of p-type a-Carbon (positive bias), the E_{in} is enhanced and the space charge region is extended. Under the 405-nm light illumination, the V_o in TiO_x converts to conductive charged oxygen vacancies (V_o^{1+} and V_o^{2+}) and photoelectrons ($V_o + h\nu \rightarrow V_o^{1+} + 1e$ or $V_o + h\nu \rightarrow V_o^{2+} + 2e$)^{24,25,30,31}. The concentration of V_o^{1+} and V_o^{2+} can be controlled by the dosage of the light stimuli, empowering the type-III heterojunction optoelectronic devices to achieve multiple memory states.

The photogenerated electrons from V_o in TiO_x are accelerated by the enhanced E_{in} . In p-type a-Carbon, electron-hole pairs are generated under light stimuli and then separated by the external electric field (E_{ext}). These holes are accelerated by the enhanced E_{in} . Consequently, the electrons from TiO_x quickly recombine with the holes from a-Carbon at the broadened space charge region. The rapid electron-hole recombination plays an essential role in constructing multiple optoelectronic memory states^{32–34}. For the n-type TiO_x , this process not only accelerates electron-vacancy separation but also decreases their recombination. For the p-type a-Carbon, this process helps the photogenerated holes rapidly transport out of the a-Carbon, avoiding the electron-hole recombination. The carrier dynamics enable stable modulation of charged oxygen vacancy concentration in the type-III heterojunction device, directly establishing multiple optoelectronic memory states. Notably, noise arising from deexcitation

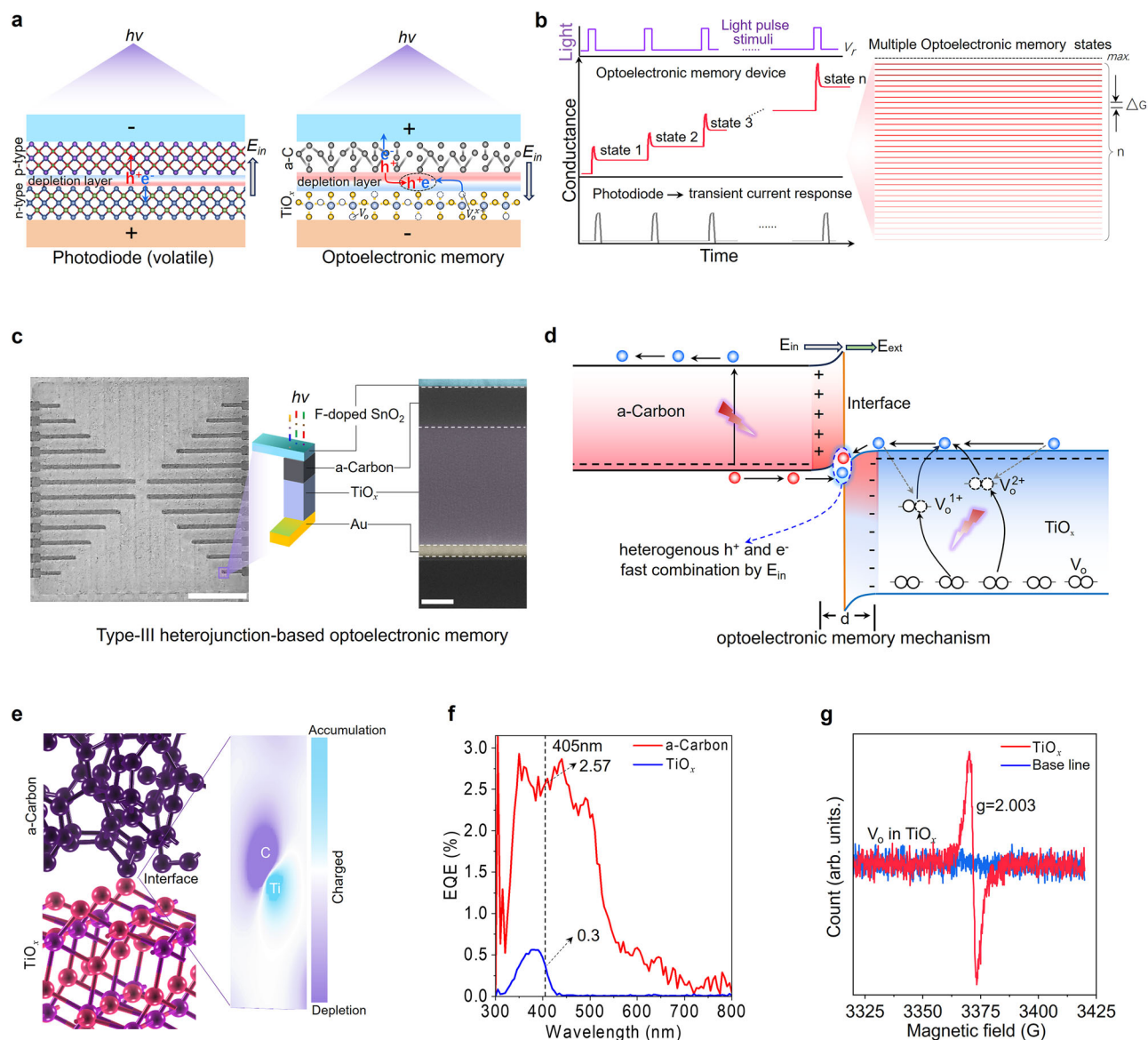


Fig. 1 | Optoelectronic memories with type-III heterojunction. **a** Schematic diagram of the photodiode (volatile) and optoelectronic memory that can store the light stimuli information. **b** Comparison of the photoresponse of type-III heterojunction optoelectronic device and that of a photodiode. The optoelectronic device can store stimuli information, but the photodiode only responds as transient photocurrents due to remarkable deexcitation recombination. The state number n is dominated by the conductance gap (ΔG) between adjacent states. **c** Type-III heterojunction optoelectronic memory with a structure of F-doped SnO_2 (~20 nm)/p-type amorphous Carbon (~120 nm)/n-type TiO_x (~350 nm)/Au (~30 nm). The heterojunction optoelectronic memory array contains 24 cells (left panel). Scale

bars: 5 μm (left), 100 nm (right). **d** Working principle of the type-III heterojunction of a-Carbon/ TiO_x , in which the deexcitation recombination of electron-hole and electron-vacancy (V_o^{1+} and V_o^{2+}) is effectively suppressed by the carrier's combination at the broadened space charge region. **e** First-principles calculations of the a-Carbon/ TiO_x heterojunction, illustrating that the electrons flow from a-Carbon to TiO_x . **f** External quantum efficiency (EQE) of the a-Carbon (red line) and TiO_x films (blue line). The EQE of a-Carbon exceeds that of TiO_x by more than eightfold at a wavelength of 405 nm. **g** Electron paramagnetic resonance of the TiO_x films (red line), indicating that the concentration of V_o , which is featured by the specific value of spectral splitting factor (g value, 2.003), is 3.10×10^{12} spins g^{-1} .

recombination (electron-hole in p-type a-Carbon and electron-vacancy in n-type TiO_x) is effectively suppressed by this carrier recombination. This design maximizes the number of encodable optoelectronic memory states within this optoelectronic device, eliminating the need for additional denoising processes.

To elucidate the deexcitation recombination suppression mechanism, we conducted first-principles calculation, external quantum efficiency (EQE) analysis, and electron paramagnetic resonance (EPR) measurement. First-principles calculations reveal unidirectional electron transfer from p-type a-Carbon to n-type TiO_x (Fig. 1e and Supplementary Fig. 13), demonstrating an inverted E_{in} orientation. This unusual behavior originates from the higher Fermi level of p-type a-

Carbon relative to n-type TiO_x . Appropriate reduction the a-Carbon thickness not only avoids overshoot photocurrent but also maintains controllable holes for the recombination with electrons from the TiO_x layer, while a relative thick TiO_x layer can accommodate more oxygen vacancy as well as reduces the leakage current.

At the wavelength of 405 nm, the EQE of the a-Carbon is more than eightfold higher than that of the TiO_x (2.57% for a-Carbon and 0.3% for TiO_x) (Fig. 1f). It suggests that p-type a-Carbon can supply much more photogenerated holes than electrons photogenerated from TiO_x . In this scenario, the quenching of photogenerated electrons is facilitated at the broadened space charge region. In consequence, the photogenerated electrons are swiftly extracted from TiO_x and

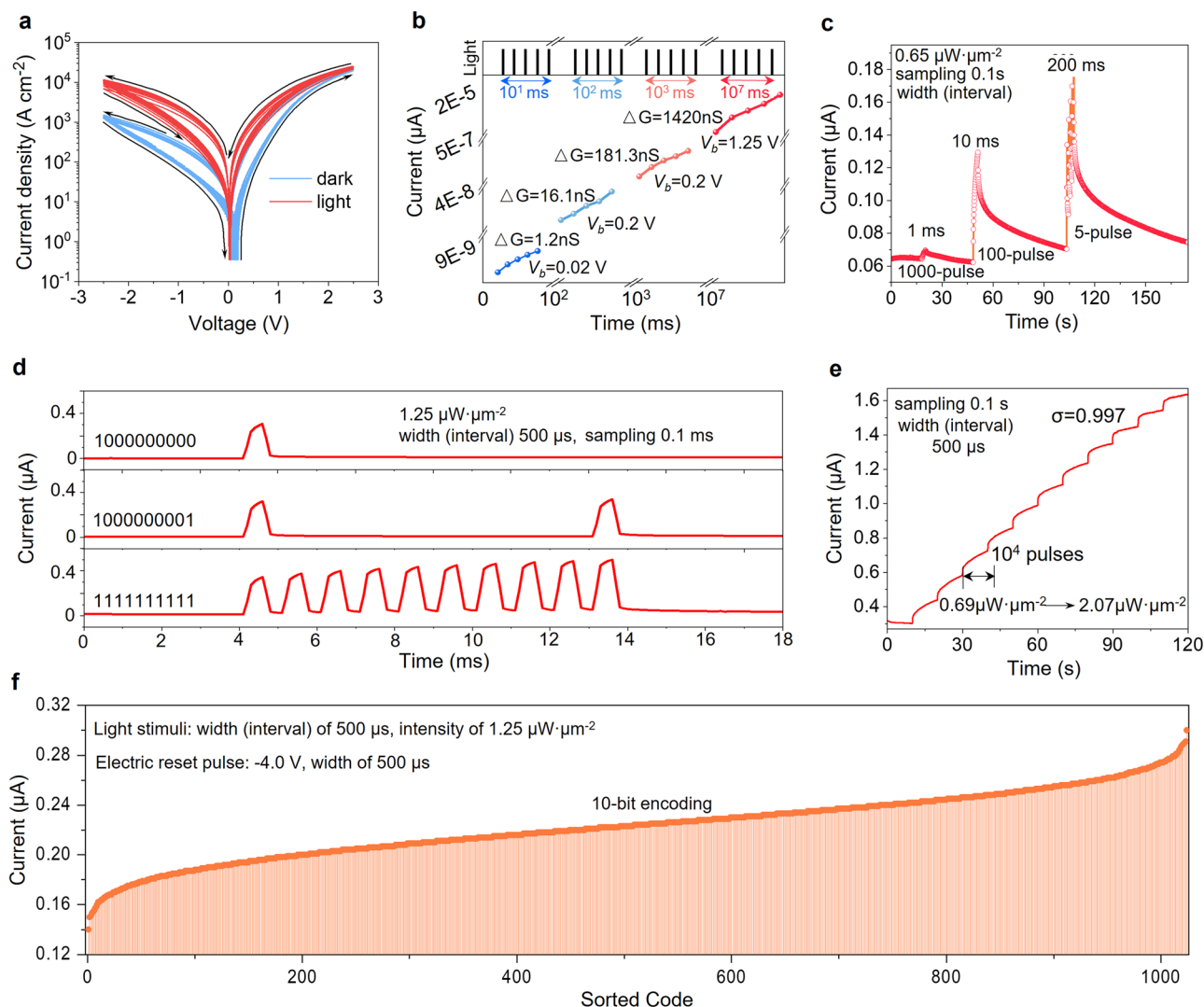


Fig. 2 | Trigger strategy of multiple optoelectronic memory states. **a** Typical current-voltage (I–V) curve of the type-III heterojunction optoelectronic memory under dark (blue line) and illumination (red line), showing an obvious positive photoresponse across the entire analog switching range. **b** Multiple states across different time scales from 10^1 to 10^7 ms. Five-pulse stimuli with intensity of $0.75 \mu\text{W} \mu\text{m}^{-2}$, width of 1.0, 10, 10^2 , 10^6 ms, and interval of 1.0, 10, 10^2 , 10^6 ms are applied to the optoelectronic device, under the matched various bias voltages (V_b) of 0.02, 0.2, 0.2, and 1.25 V, respectively. **c** Photoresponse under light stimuli with width (interval) of 1.0, 10, and 200 ms and intensity of $0.65 \mu\text{W} \mu\text{m}^{-2}$. The 1000, 100, and 5 pulses are respectively operated for each type light stimuli, indicating that small

conductance gap (ΔG) between two adjacent states can be established by narrow light pulse. **d** The photoresponse of “1000000000”, “1000000001”, and “1111111111” pulse chain triggered by light stimuli with an intensity of $1.25 \mu\text{W} \mu\text{m}^{-2}$ and width (interval) of 500 μs (50% duty ratio). The photoresponse under $1.25 \mu\text{W} \mu\text{m}^{-2}$ and dark is defined as “1” and “0”, respectively. **e** Linear modulation of photoconductance. The conductance is progressively modulated by the light stimuli with width (interval) of 500 μs and intensities ranged from 0.69 to $2.07 \mu\text{W} \mu\text{m}^{-2}$, showing a high linearity of 99.7%. The ten thousand of pulses are operated for per light intensity. **f** 10-bit (1024 distinguishable states) encoding photoresponse under the light stimuli (intensity of $1.25 \mu\text{W} \mu\text{m}^{-2}$, width of 500 μs , interval of 500 μs).

eliminated in the junction region, in turn suppressing deexcitation recombination of the V_o^{1+} (or V_o^{2+}) with photogenerated electrons in TiO_x . This process in-situ reduces noise from de-excitation recombination during the programming of optoelectronic devices. The optical band gap of 3.16 eV in TiO_x ensures that nearly all photogenerated electrons originate from oxygen vacancy sites rather than its valence band, although it exhibits a low EQE value.

The existence of V_o is characterized by the EPR. The V_o is featured by a specific spectral splitting factor (g value) of 2.003, and its concentration is 3.10×10^{12} spins g^{-1} (Fig. 1g and Supplementary Fig. 14). Additionally, we extracted the threshold voltage (V_th , 1.15 V) from the space charge limited current mechanism fitted by double logarithm I–V curves of the $\text{Au}/\text{TiO}_\text{x}/\text{Au}$ device, showing a V_o concentration of $6.6 \times 10^{22} \text{ m}^{-3}$ (Supplementary Fig. 15). These oxygen vacancies are the source of the optoelectronic memory states. Under light illumination,

electrons are detrapped from the neutral oxygen vacancy sites, converting the oxygen vacancy from neutral state to charged state. The optoelectronic memory states are reconfigured by controlling the concentration ratio of the charged oxygen vacancies.

High-precision programming for thousands of states

The current-voltage (I–V) curves measured under both dark and illumination reveal that the optoelectronic memory exhibits a positive photoresponse across the entire range of its analog switching behaviour (Fig. 2a). For electric switching mechanism, electrons are progressively injected into the a-Carbon layer when positive voltage scan is applied on Au, subsequently traversing the narrowed space charge region to generates a continuous increase current. When negative voltage is operated, the electron injection is significantly suppressed in TiO_x due to scattering effect and Coulomb blockade effect. The

continuous electrical switching is dominated by the electron trapping/detrapping dynamics instead of the filamentary conduction mechanisms^{8,17,32}.

The photoresponse depends on the light dosage and can be efficiently modulated by pulse count, frequency, width, and light intensity (Supplementary Fig. 16a–e). The optoelectronic memory under two consecutive light pulses with different intervals exhibits typical paired-pulse facilitation, suggesting the good potential to emulate retina synapses for vision information processing (Supplementary Fig. 16f).

The V_b builds E_{ext} that can affect the E_{in} amplitude. Under positive bias condition, a large V_b accelerates carriers fast-transport and rapidly recombines, thus strengthening the suppression of the deexcitation recombination. Under specified V_b and light stimuli with fixed duty ratio and fixed intensity, the high-frequency stimuli (and correspondingly lower light dosage per pulse) generate smaller ΔG values, which allows the optoelectronic memory to build more states within its intrinsic conductance range.

Different ΔG (1.2; 16.12; 181.3; and 1420 nS) can be respectively triggered by the stimuli (0.02 V, width of 1.0 ms, interval of 1.0 ms; 0.2 V, width of 10 ms, interval of 10 ms; 0.2 V, width of 10^2 ms, interval of 10^2 ms; and 1.25 V, width of 10^6 ms, interval of 10^6 ms) with constant intensity of $0.75 \mu\text{W} \mu\text{m}^{-2}$ (Fig. 2b). After light stimuli, the optoelectronic memory states can be well maintained for 1000 s, suggesting that these optoelectronic memory states that across different time scale are nonvolatile (Supplementary Fig. 17). These optoelectronic memory states can be well modulated and retained in different time scales, allowing for various perception tasks across various time scales.

Ultraviolet light with wavelength of 405 nm, intensity of $0.65 \mu\text{W} \mu\text{m}^{-2}$, and width (interval) of 1, 10, and 200 ms is applied to the type-III heterojunction optoelectronic memory device, yielding the ΔG values of 0.01, 0.58, and 19.80 nS (Fig. 2c, 0.1 s sampling step). The finer photocurrent response is feasible for higher light stimuli frequency ($2.5 \mu\text{W} \mu\text{m}^{-2}$, width of 50 μs , interval of 50 μs) (Supplementary Fig. 18). Therefore, the ΔG of the type-III heterojunction optoelectronic memory simultaneously relies on the light stimuli condition (width, interval, and intensity) and the bias, showing as increase as light dosage and bias voltage (Supplementary Fig. 19).

For the type-III heterojunction optoelectronic memory, the non-volatile memory states are stably maintained through optical programming at “1” positions, while active electrical modulation at “0” positions enable fine-tuned control via light pulse sequence-dependent adjustments. Figure 2d demonstrates the encoding results of “1000000000”, “1000000001”, and “1111111111”. The “1” represents the photoresponse under one pulse light stimulus (intensity of $1.25 \mu\text{W} \mu\text{m}^{-2}$, width of 500 μs , interval of 500 μs , and 0.1 ms sampling step) while the “0” position is matched a reading voltage pulse (0.2 V, 500 μs) denotes the photoresponse under darkness (Supplementary Fig. 20). Under 1000 Hz light stimuli (width of 500 μs , interval of 500 μs), the photoresponse is progressively adjusted by various intensities from 0.69 to $2.07 \mu\text{W} \mu\text{m}^{-2}$, demonstrating a high linearity of 99.7% (Fig. 2e, 0.1 s sampling step). This high linearity enables the optoelectronic memory to establish uniform ΔG across multiple memory states, ensuring precise and reliable optical information encoding.

With the multi-bit encoding strategy, different pulse chains yield distinct optoelectronic memory states (Supplementary Fig. 21). Figure 2f is the sorted photocurrent for the 10-bit encode, in which each current value is achieved after the light stimuli with an intensity of $1.25 \mu\text{W} \mu\text{m}^{-2}$ and width (interval) of 500 μs . Therefore, the type-III heterojunction optoelectronic memory can provide thousands of distinct states for fast vision encoding under specific light dosage by appropriate light stimuli.

Thousands of states in the optoelectronic memory

Figure 3 shows the thousands of states achieved from the type-III heterojunction optoelectronic memory. Left inset of Fig. 3a

schematically shows the optoelectronic memory states triggered by the light pulse chain according to the 10-bit binary code. According to the 10-bit binary code chains, 1024 optoelectronic memory states are achieved after operating the corresponding light pulse chains with intensity of $1.25 \mu\text{W} \mu\text{m}^{-2}$, width of 500 μs , and interval of 500 μs (right inset of Fig. 3a). After light stimuli for each memory state, the type-III heterojunction optoelectronic memory is reset by a negative voltage pulse (−4.0 V, 500 μs). The 1024 optoelectronic memory states only possess a volatility rate of ~3% within 1000 s (Supplementary Fig. 22a). The enlarged region of the Fig. 3a illustrates that these optoelectronic memory states, which are one-by-one triggered by the 10-bit light pulse chain, can be completely distinguished (Supplementary Fig. 22b). To verify the reliability of these states, we carried out the current fluctuation analysis. Figure 3b depicts the temporal evolution of the programmed current for 420th and 421st states under 0.2 V. The statistical analyses of the 420th and 421st states exhibit a fluctuation of ~0.19 nA, which is smaller than the average current gap ($\Delta I = 0.245$ nA) between adjacent levels among the 1024 states. The programmed conductance states can be well distinguished (Fig. 3c). The mean value gap of 0.25 nA demonstrates a clear separation between the 420th and 421st state. The statistics for the 420th and 421st states, based on consecutive sampling points (sample n and sample $n + 1$), reveal tight intra-state distributions and clear inter-state separation (Fig. 3d). The 1024 states are tested repeatedly on the same cell 50 times (Fig. 3e). Each set of the 1024 states is highly consistent and fully reproducible. Therefore, the type-III heterojunction optoelectronic memory shows good cycle-to-cycle stability for the 1024 states. After each programming cycle, the optoelectronic memory can be reset by a negative voltage (−4.0 V, 500 μs). The 1024 states are available for each cell within the 24 cells, suggesting excellent device-to-device stability (Supplementary Fig. 23). Figure 3f exhibits the average gap between adjacent states (ΔG) and the current range (i.e., the difference between the 1st state and the 1024th state), highlighting both cycle-to-cycle and device-to-device stability. These stability metrics illustrate that the 1024 states are distinguishable with a ΔG of 1.27 ± 0.02 nS and a current range of 0.28 ± 0.015 μA across 50 cycles and 24 cells. In comparison, the type-II heterojunction optoelectronic memory (FTO/a-Carbon/MoS₂/Au) and the control optoelectronic memory (Au/TiO_x/Au) show more than two-order fluctuation (~200 nS) (Supplementary Figs. 24–26).

This photoelectronic synergy encoding mechanism fundamentally relies on the nonequilibrium dynamics of photogenerated carriers. In the a-Carbon layer, intense light absorption generates a substantial population of photogenerated electrons and holes, resulting in an overshoot photocurrent due to carrier accumulation. Conversely, in the TiO_x layer, photogenerated electrons originate from neutral oxygen vacancies (V_o), which subsequently transform into charged oxygen vacancies (V_o^{x+}), a process that underpins nonvolatile photoconductance memory states.

The recombination of majority carriers (holes, h^+ , from a-Carbon) with minority carriers (electrons, e^- , from V_o) in the space charge region leads to a surplus of h^+ within this region. Driven by the built-in electric field (E_{in}), these surplus h^+ migrate toward the TiO_x side, effectively shortening the space charge region. This carrier redistribution, however, diminishes the width of the space charge region, thereby weakening its efficiency.

To counteract this weakening effect, a matched positive bias of +0.2 V (E_{ext}) is applied to the “0” position. This external field compensates for the E_{in} reduction caused by carrier nonequilibrium, restoring the space charge region width and barrier integrity. Importantly, it introduces timing sequence sensitivity, enabling precise light pulse modulation. Consequently, the programming process achieves enhanced stability and reproducibility, as the bias actively mitigates carrier-induced nonequilibrium effects while leveraging the pulse sequence for dynamic control.

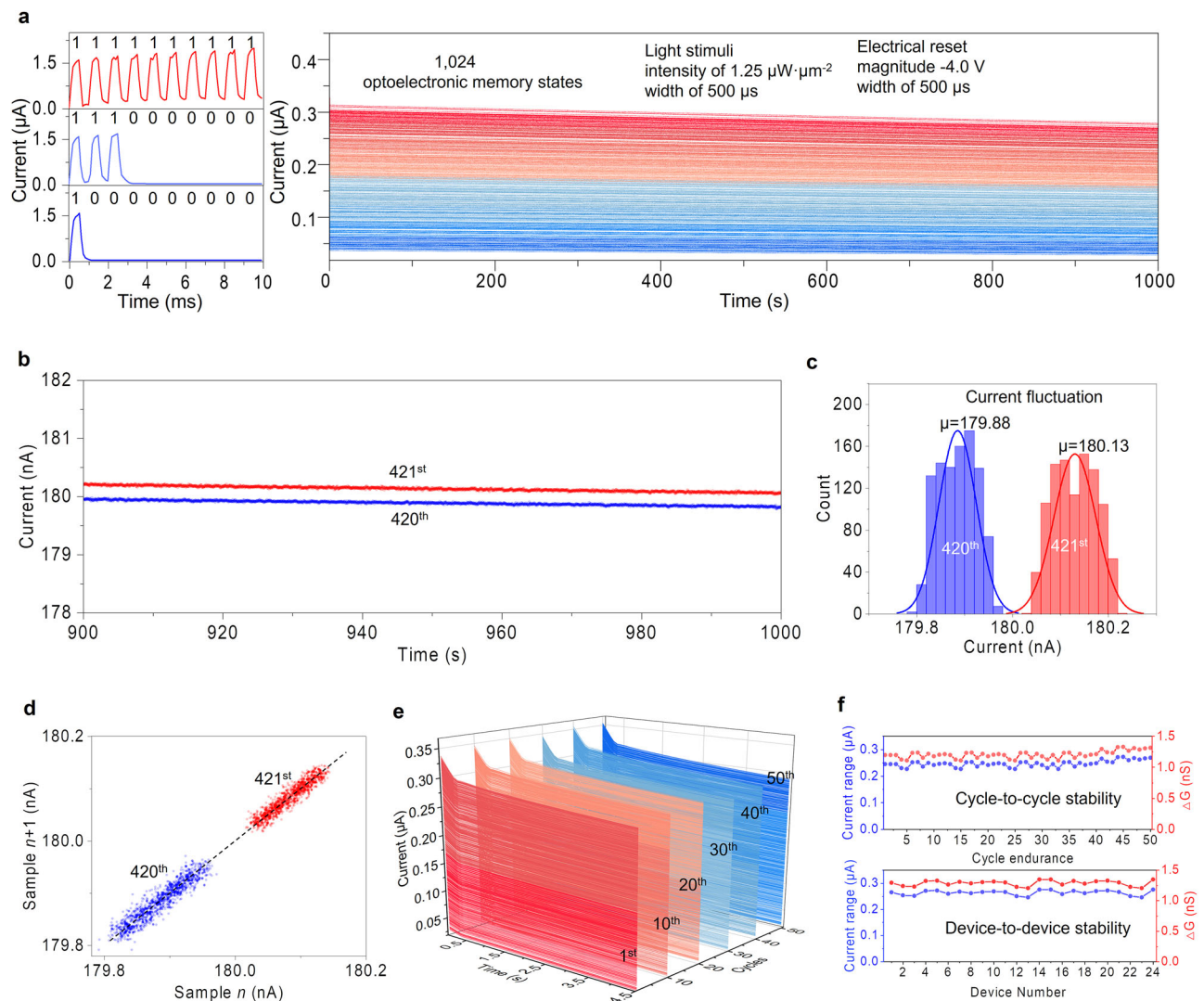


Fig. 3 | Thousands of optoelectronic memory states in the type-III heterojunction device. **a** 1024 optoelectronic memory states achieved from the type-III heterojunction optoelectronic memory. The left inset schematically shows the optoelectronic memory states triggered by the light pulse chain according to the 10-bit binary codes in which the “1” and “0” represents with light illuminating and darkness. The light stimuli possess width of $500 \mu\text{s}$, interval of $500 \mu\text{s}$, and intensity of $1.25 \mu\text{W} \mu\text{m}^{-2}$. The matched read voltage is 0.2 V . **b** The programmed current over time for the 420th (blue line) and 421st (red line) states between 900 and 1000 s under 0.2 V reading. **c** The statistical analysis of current sampling points of the 420th (blue line) and 421st (red line) states between 900 and 1000 s, showing a

minor fluctuation of -0.19 nA for each state. The mean current gap of -0.25 nA (the mean (μ) difference between two programmed states) exceeds the fluctuation of -0.19 nA , indicating that the two states are well distinguishable. **d** The statistical results of the sample n and $n+1$ (n and $n+1$ refer to consecutive sampling points on the time-resolved current curve) for the 420th (blue line) and 421st (red line) states, demonstrating the compact distribution within each state. The distributions of the 420th and 421st states are distinctly separable. **e** The 1st–50th cycling (from red to blue) test operated on a randomly selected cell from the array, demonstrating good reproducibility. **f** The average ΔG (red colour) and the current range (blue colour) of the 1,024 states, highlighting the device-to-device and cycle-to-cycle stability.

The inter-physical mechanism enables the designed type-III heterojunction optoelectronic device with thousands of optoelectronic memory states, but the number of the distinguishable states may be affected by noise sources such as deexcitation recombination noise, RTN, shot noise, and thermal noise. The shot noise can play a significant role in photodiodes operating under reverse bias^{35,36}. However, at room temperature and within the terahertz range, thermal and shot noise are frequency-independent and have minimal impact on devices operating at low frequencies²¹. In our device, the programmed optoelectronic memory state is read under a direct current (DC) forward bias. The de-excitation recombination noise and RTN as the possible primary noise sources. RTN can be attributed to either electronic³⁷ or atomic³⁸ effects.

By introducing EQE difference across the type-III heterojunction, deexcitation recombination noise is in-situ suppressed during the

optical programming of the proposed optoelectronic memory. The proposed device is programmed optically instead of electrically, preventing defect migration in the switching layer. This precludes atomic effect-induced RTN. The optical switching behaviours in the proposed device relies on the electron detrapping from oxygen vacancies and collective hopping. RTNs from electronic effects are intimately associated with carrier trapping/emission at local defects. According to the Poole-Frenkel model³⁹, an applied electric field reduces the emission barrier for electrons in the defect and increases the local temperature nearby, thereby elevating the rate of electron capture and emission at the defect⁴⁰. Furthermore, the conductance modulation in the proposed device primarily relies on defects (V_o) near the heterojunction interface. The band bending at the heterojunction may further accelerate carrier capture and emission at these defects. The greatly accelerated carrier dynamics result in smoother level averaging and

reduced RTN amplitude. Moreover, carrier hopping in the devices involves many defects. The superposition of RTN effects across these defects makes RTN from a single defect indistinguishable, thereby diminishing its impact⁴¹. The low noise amplitude after programming in the proposed device confirms the minimal impact of RTN. Additionally, the sum of multiple random variables tends to follow a normal distribution, provided each variable's influence is limited⁴². In the proposed device, the individual states after programming follow a normal distribution (Fig. 3c), indicating that RTNs from multiple defects overlap, thereby reducing their impact.

The RTN in oxide memristors comes from the incomplete conduction filaments (ICM)¹⁷, while electron trapping and detrapping along with collective hopping effect eliminate the filament formation in the type-III heterojunction optoelectronic memory^{40,41}. The electric fast-RESET (~ 2.5 V, 100 μ s), continuous electrical switching of the control samples (FTO/a-Carbon/Au, FTO/TiO_x/Au), and SA-TEM observation confirm that the type-III heterojunction optoelectronic memory is dominated by electron trapping and detrapping mechanism instead of the conduction filaments that generally cause an abrupt SET and RESET processes^{43–46} (Supplementary Figs. 27–29). In addition, the optoelectronic memory states with small fluctuation (~ 1.0 nS) at different temperatures (300–350 K), indicating that thermal noise is negligible (Supplementary Fig. 30).

High-precision spatiotemporal fusion for motion perception

The proposed device possesses thousands of optoelectronic memory states and a high signal-noise ratio of 51.0 dB and high information transmission rate of 1.7×10^4 bit s⁻¹ under light stimuli (intensity of 2.5 μ W μ m⁻², width (interval) of 0–500 μ s), allowing to faithfully emulate the raptor vision for fast-moving object perception (Fig. 4a and Supplementary Fig. 31). It integrates three key functions: highly sensitive and rapid conversion the light information, swift integration the information, and graded information processing (Fig. 4b). The a-Carbon emulates the high light sensitivity and swift information transmission of the dual cones with an accessory member and a principal member with an oil droplet that is highly sensitive to light and rapidly execute light-to-electronic conversion^{47,48}. The type-III heterojunction collects the electron and hole and then fast combine them as the manner of information integration of the horizontal cells. The TiO_x layer offers optically programmable vacancies for information grading processing as the manner of bipolar cells and others.

Figure 4c exhibits an architecture of the raptor vision-inspired perception and recognition system. The CMOS-based time-integrated exposure or multi-exposure photography has to rely the complex circuit shutter and then fused the frame one by one⁴⁹. In contrast, our device provides high-precision frame-free information fusion strategy (Supplementary Fig. 32). A full hardware system is developed (Fig. 4d). The front terminal subsystem executes 10-bit spatiotemporal information fusion for motion perception. An electrical RESET (~ 4.0 V, width of 500 μ s) is operated after 10-bit encoding. The end-terminal subsystem executes the CNN inference for image recognition (Supplementary Figs. 33 and 34).

Figure 4e, f demonstrate the raptor vision-inspired hardware system for real wood block moving perception and recognition. The moving from t_0 to t_n is perceived by the front terminal subsystem. The moving action is well classified with 100% accuracy. Complex unmanned aerial vehicle (UAV) with a high speed (>180 km h⁻¹) can be precepted when the front terminal subsystem executes 10-bit encoding (Fig. 4g). And it shows an accuracy of 100% and 98.75% for five UAV actions (five categories: up, down, left, right, and blank) and seven UAV actions (seven categories: up, down, left, right, approaching, leaving, and blank) with complex flight postures, including accelerating, decelerating, rising, falling, reciprocating, and circling (Fig. 4h and Supplementary Fig. 35). Therefore, the thousands of optoelectronic

memory states outweighs other optoelectronic device capability to fuse the spatiotemporal information (Supplementary Table 1).

Compared with conventional electrical operation systems (e.g., CPUs, FPGAs, GPUs), the spatio-temporal encoding system leveraging type-III heterojunction optoelectronic memory demonstrates a power consumption reduction of at least one order of magnitude (Supplementary Table 2). This efficiency gain stems from the system's ability to perform optical programming and electrical modulation in a synergistic manner, minimizing energy-intensive electrical switching.

Furthermore, the system achieves a programming throughput of 24,000-bit s⁻¹ when executing 10-bit parallel encoding across 24 memory cells within a 10 ms time window. This performance highlights the scalability of the optoelectronic architecture, as the throughput scales linearly with the number of cells, enabling high-speed data processing without compromising energy efficiency.

A type-III heterojunction optoelectronic memory can provide thousands of optoelectronic memory states for highly precise perception. The photogenerated carriers (holes from p-type a-Carbon and electrons from n-type TiO_x) quickly combine at the broadened space charge region. This process efficiently dampens the deexcitation recombination (major noise source) of electron-hole in a-Carbon and that of electron-charged oxygen vacancy in TiO_x. Without using post-programming denoising, 1024 optoelectronic memory states are constructed. The distinguished thousands of memory states enable the type-III heterojunction optoelectronic memory to hold a high SNR of 51.0 dB at 1000 Hz and an information transmission rate of 1.7×10^4 bit s⁻¹ within 0–1000 Hz bandwidth. This work provides a significant optoelectronic mechanism to manufacture denoising-free optoelectronic memory element for highly precise spatiotemporal information fusion.

Methods

Device fabrication

The optoelectronic memory devices were fabricated by radio frequency (RF) magnetron sputtering technology and physical vapor deposition. An amorphous Carbon (a-Carbon) target with a high purity of 99.999% was sputtered in 1.0 Pa Ar ambient at the power of 30 W for 30 min. For the TiO_x layer, 100 W sputtering power and 150 V bias voltage were applied on a TiO₂ target with a purity of 99.99% for 60 min under the 1.0 Pa high-purity Ar ambient (purity $>99.99\%$). The F-doped SnO₂ (FTO) electrodes consisting of 32 lateral bars were prepared by sputtering while the Au top electrodes were deposited by physical vapor deposition under ultra-vacuum conditions (5.0×10^{-4} Pa). According to the above methods, a device with a sandwich structure (FTO/a-Carbon/TiO_x/Au) and a corresponding crossbar array with a scale of 32×32 was developed in this work.

Optoelectronic property measurements

An integrated electrical test system composed of semiconductor analyzer systems (Keithley 4200-SCS, B1500A) and probe station (Lakeshore, TTPX) was employed to conduct the electrical measurements, such as the current-time (*i*-*t*) and current-voltage (*I*-*V*) tests. An integrated photoelectrical test system, including a 405-nm laser (Cobolt, 0405-06-01-0100-100) with tunable light intensity, a laser (Chang Chun Liang Li optoelectronic Equipment Co., Ltd.) with tunable light wavelength (this work uses the wavelength of 405 nm) and light intensity, and function waveform generator (UTG7062B) for light frequency modulation, is used to perform the optoelectrical measurements of the FTO/a-Carbon/TiO_x/Au devices. Moving objects are generally described by the fusion of sequential 4–10 frames. Thus, sequential light stimuli with 4–10 pulses could be applied to the heterostructure optoelectronic devices for mapping the spatiotemporal information of the moving objects in the real world. Each of the optoelectronic devices can independently sense and process the

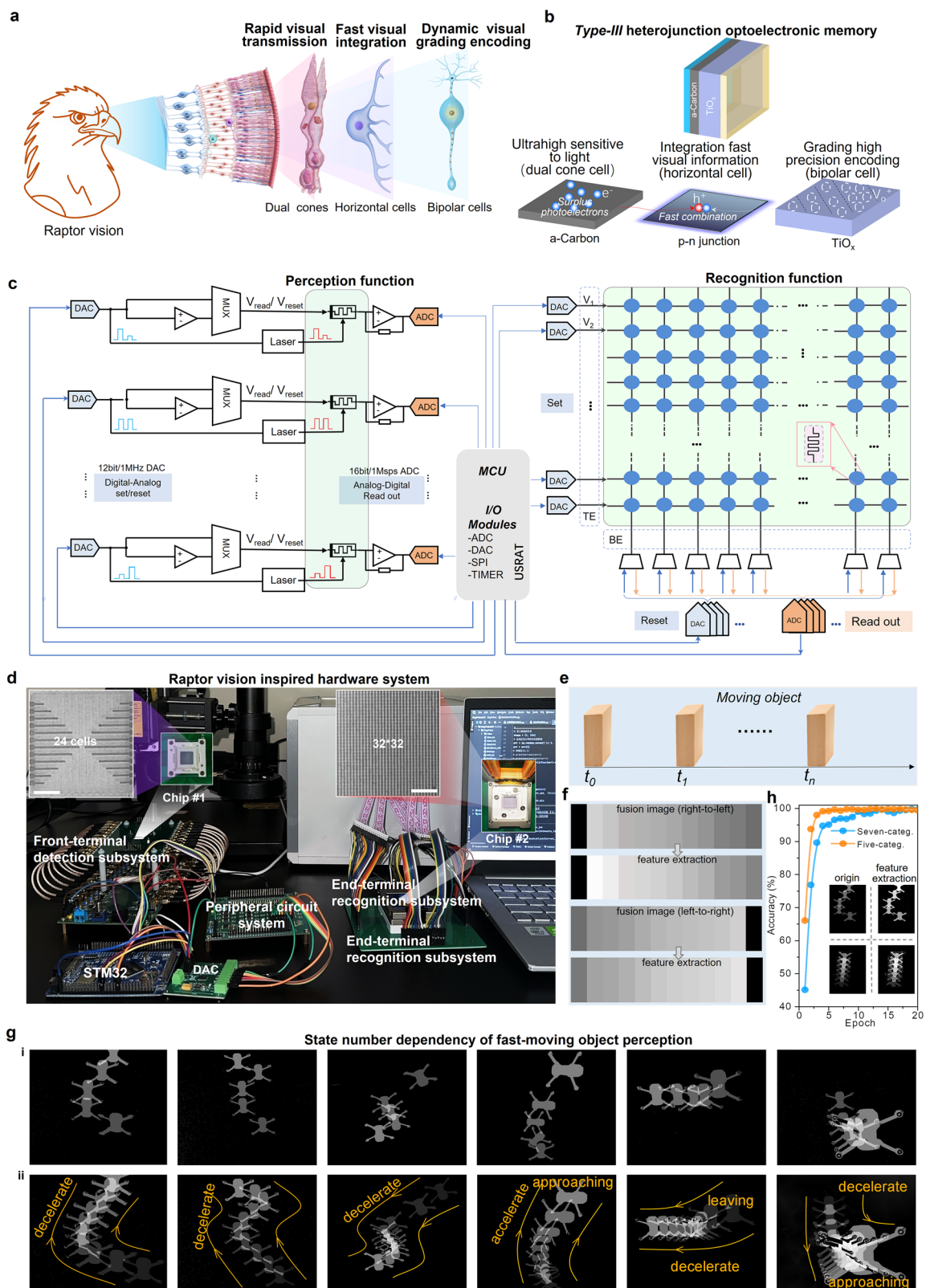


Fig. 4 | Spatiotemporal information fusion hardware system. **a** Schematic diagram of raptor retina system. The dual cone cell consists of an accessory member and a principal member with oil droplet. The horizontal cell and bipolar cell rapidly integrate visual information. **b** Raptor vision-inspired optoelectronic memory. **c** Architecture of the raptor vision-inspired perception and recognition system. **d** Hardware of high-precision spatiotemporal information fusion and

recognition system. Scale bar: 5 mm. **e, f** Demonstration of the hardware system for moving wood block perception and recognition. **g** Perception simulation of complex fast-moving ($>180 \text{ km h}^{-1}$) unmanned aerial vehicle (UAV). **h** Recognition of five (orange colour) and seven (blue colour) UAV actions, showing an accuracy of 100% and 99.5%.

spatiotemporal frame information at the pixel level based on the temporal accumulating photocurrents under a series of light pulses. Accordingly, the photocurrent response under different sampling frequencies (the frequency of light pulse application used to capture information about the motion of a moving object) ranging from 10^{-4} – 10^4 Hz was used to capture the action information of the moving objects with varying speeds.

Electron state calculations

The (101) surface of anatase TiO_2 and the a-Carbon from Timothy Clark's study were built to investigate the electron states at the a-Carbon/ TiO_x interface based on the density functional theory, which was implemented in the Cambridge Sequential Total Energy Package. To avoid mirror interactions, we employed a surface model with a 20 Å vacuum thickness on the (101) tetragonal structure. The structure optimization utilized a plane-wave cutoff energy of 750 eV and Monkhorst–Pack k-point sampling of the Brillouin zone with Gamma points. The convergence tolerance for energy during geometric optimization was set to 3×10^{-5} eV atom $^{-1}$. The residual force was 0.00089 eV Å $^{-1}$, and the maximum atomic displacement was 0.00095.

Neuromorphic computing simulations

We demonstrate action detection and recognition of complex moving objects in the real world. A comparison of the ability to detect and recognize complex actions between the raptor retina-inspired optoelectronic memory device and conventional image sensors was made. A dataset composed of 2800 images was customized based on the UAV flight video that includes 7 types of actions (up, down, left, right, approaching, leaving, and blank). The single-frame image with a scale of 32×32 pixels was mapped as “1” and “0” in the gray level, suggesting that the specific space information could be obtained from the single-frame image. Thus, the detection of the complex actions of the moving objects based on multi-frame fusing images with spatiotemporal information can be obtained through mapping the specific pixel information on the raptor retina-inspired optoelectronic memory devices. The recognition system was built using a convolutional neural network (CNN) consisting of an input layer, a $3 \times 3 \times 4$ convolutional kernel, a 2×2 subsampling pooling. ARM processor is used to operate the ReLU activation and 2×2 max pooling (stride 2), compressing the data into a $3 \times 3 \times 1$ feature vector (9-dimensional). This vector feeds into a 9×7 fully-connected layer to generate a 7-dimensional output, with final classification probabilities delivered via the SoftMax function.

Data availability

The datasets generated in this study have been deposited in the Figshare database under accession code <https://doi.org/10.6084/m9.figshare.31649767>.

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Author contributions

Y.C. and G.Z. conceived the concept and supervised the project. G.Z. conducted the device design, fabrication, and measurements. G.Z., Y.X., and C.Z. analyzed the experimental data. X.X. and C.Z. built the algorithm and simulation. G.Z. and J.Y. conducted the theory calculation. G.Z., Y.X., and C.Z. wrote the manuscript and all authors contributed to the revision of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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